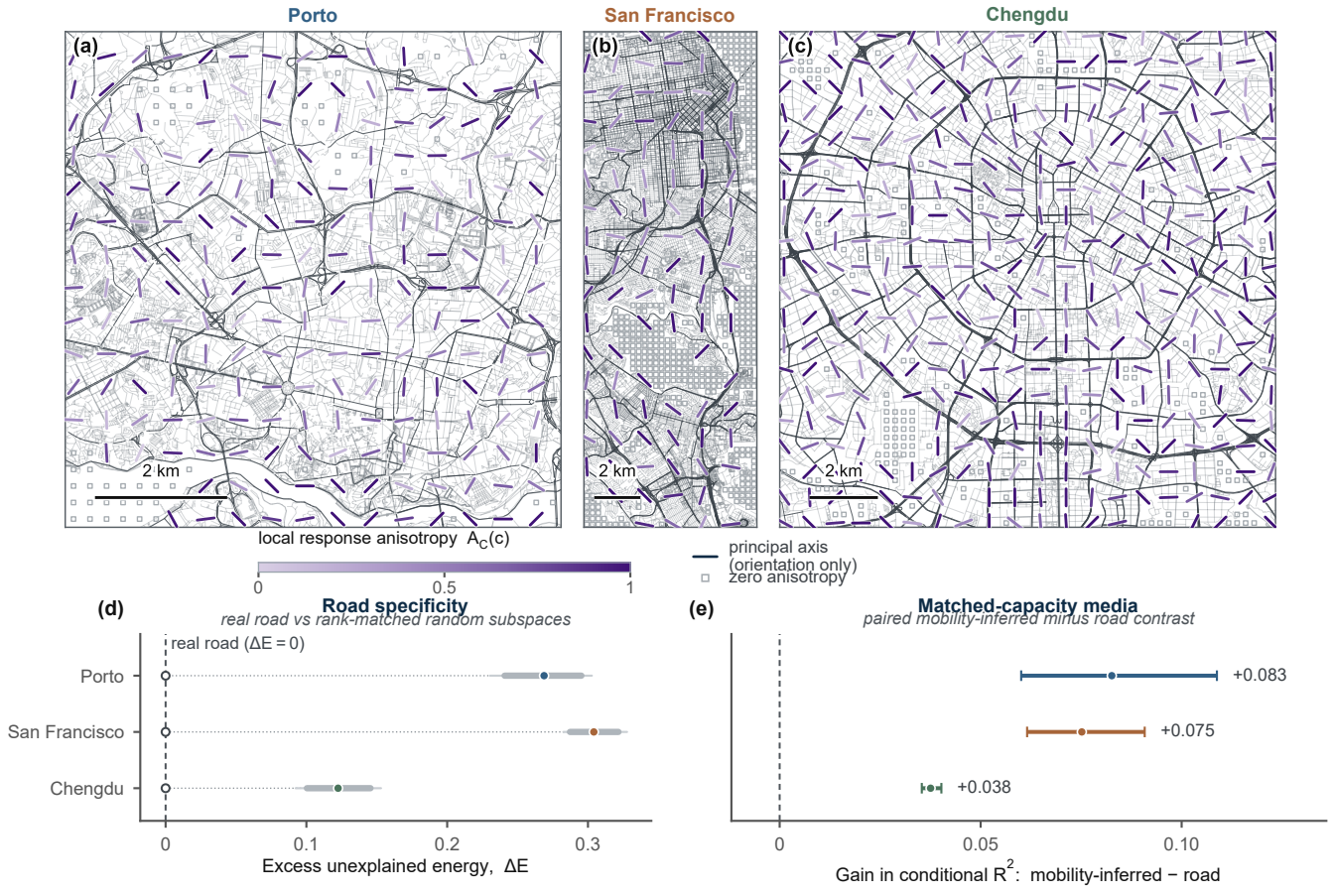

Porto
San Francisco
Chengdu
(a)
(b)
(c)
2 km
2 km
2 km
local response anisotropy $A_C(c)$
0
0.5
1
principal axis
(orientation only)
zero anisotropy
(d)
Road specificity
real road vs rank-matched random subspaces
real road (ΔE = 0)
Porto
San Francisco
Chengdu
0
0.1
0.2
0.3
Excess unexplained energy, ΔE
(e)
Matched-capacity media
paired mobility-inferred minus road contrast
+0.083
+0.075
+0.038
0
0.05
0.10
Gain in conditional R^2: mobility-inferred − road